Research Talk

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It mainly focus on developing rigorous, computational efficient and generalizable Statistical Methodology to understand the change of data - specifically in the deployed Machine Learning(ML) systems.
Main Scientific & Practical Questions to address:

- Given two batches of data $D_{exist} = (\mathbf{X}_{exist}, Y_{exist})$ and $D_{new} = (\mathbf{X}_{new}, Y_{new})$. How to know whether the difference is from concept drift($P(Y|\mathbf{X})$) or covariate shift($P(\mathbf{X})$)?
- How to understand the change of data from the feature level?
- Given a deployed ML model in the production system, how to know when and how should we update it?

Overview of Methods

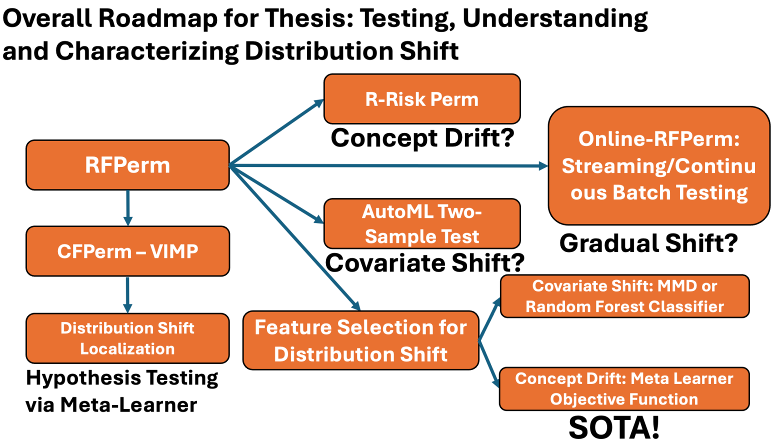


Figure: Overview of the Methodologies we have developed.

Methodology 1 - Random Forest Permutation Test

Permutation test to compare validation error(OOB Error) and test error(Prediction Error) in Random Forest.

- Random Forest, well known for its robustness across a wide variety of different scenarios, would be the best choice to isolate the effect of OOD(out-of-distribution).
- It would be capable of characterizing both **Concept Drift** and **Covariate Shift** – Density Aware nature of Random Forest.
- Generalizable to arbitrary Machine Learning Models on arbitrary Modality of data - as long as the validation set is well-defined. The software is open-sourced in `https://github.com/ruanhq-StatsML/RFPerm`.

Methodology 2 - Meta-Learner based Hypothesis Testing

Leverage Meta-Learners to detect Distribution Shift - via Variable Importance. D_{exist} as Control Batch($T = 0$), D_{new} as Treatment Batch($T = 1$).

- Permute-then-refit the Causal Learner, looking at the variable importance(LOCO or permutation variable importance).
- The variable importance in this framework itself can be interpreted as the quantification of the contribution to certain types of distribution shift, potentially OOD variable importance.
- The meta-learner serves as a proxy as the distance estimator instead of causal estimator. The software is open-sourced in `https://github.com/ruanhq-StatsML/Causal_Objective_Permutation_Test`.

Meta-Learner Testing for Feature Selection and Distribution Shift Localization

The proposed meta-learner hypothesis testing framework (**CFPerm**) inherently serves a dual purpose:

1. Identifying informative features driving distributional changes.
2. Enabling subset localization of the sources of shift.

- **Decision Rule**: A feature is flagged as a significant driver if its original Variable Importance (VIMP) score is significantly larger than its permuted VIMP distribution under the null hypothesis.
- **Subset Localization**: The selected significant features not only reveal the primary drivers of the distribution shift, but the distinct subsets they induce also serve as highly efficient proxies for pinpointing where the shift is most pronounced.
  - *Insight*: This facilitates interpretable diagnostics by directly targeting high-drift regions in the feature space.

Leverage Meta-Learners, R-learner to develop hypothesis testing procedure for isolating concept drift from covariate shift(Causal objetive function)

- The motivation of R-learner is similar to that in the Double/Debiased Machine Learning(DML) - Remove the effect of the covariates.
- Based on Projection Theory, $\tilde{Y} = Y - m(X) = (I - \Pi_X)(Y)$ and $\tilde{T} = T - e(X) = (I - \Pi_X)T$. The R-risk is invariant under Covariate Shift, isolating of the Concept Drift, serve as an objective function for isolating the concept drift.
- Can be generalizable to arbitrary Objective Functions in literatures of Causal Inference.

Methodology 3 - Feature Selection for Distribution Shift

In production, to conduct feature-level attribution, the following aspects can be leveraged:

- For Covariate Shift only, MMD+LOCO(leave-one-covariate-out) or train a random forest to distinguish two datasets is efficient.
- For Concept Drift, leveraging the Meta-Learner(DR-learner with Permutation VIMP) procedure for Feature Selection.
- Proactive and Actionable diagnostic tools for feature quality deterioration - feature selection in near real-time, enabling tractable root-cause analysis.

Methodology 4 - Online Extension of RFPerm

Suppose a deployed model $\hat{f}$ is launched in Production. In the streaming data setting, new observations $\cup_{i=1}^{n}\{\mathbf{X}_i, Y_i\}$ coming continuously, we aim to give people an idea of when the gradual distribution shift become significant - trigger model update.

- A procedure based on empirical p-values is proposed to assess the extremity of the current predictive error relative to those in the deployed model.
- An online False Discovery Rate (FDR) control procedure is incorporated to effectively mitigate false alarms.
- **Model Agnostic** as long as a valid predictive model is well-defined and near-real-time inference is enabled.
- The software is available at `https://github.com/ruanhq-StatsML/OnlineRFPerm`.

Methodology 4: Illustration

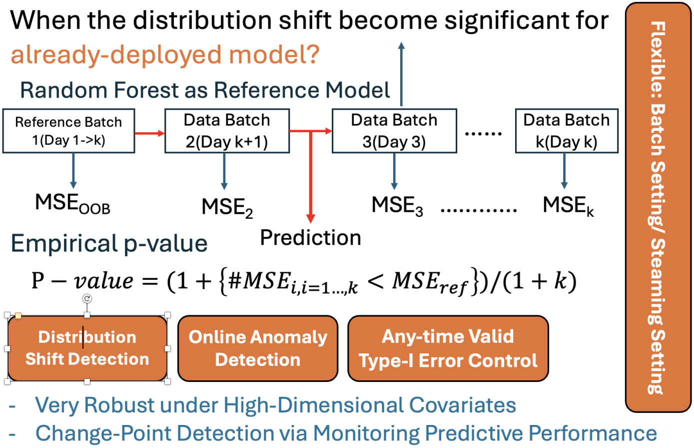
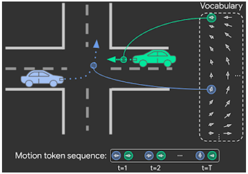


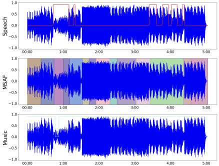
Figure: Streaming Testing for Gradual Distribution

Methodology 4: Flexibility and Modality Agnostic

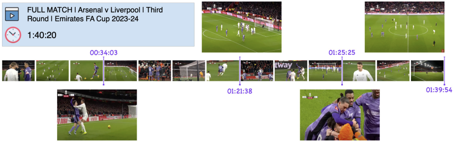
The methodology is model-agnostic, with demonstrated effectiveness across diverse data modalities—making it broadly applicable to any ML system without architectural modification.



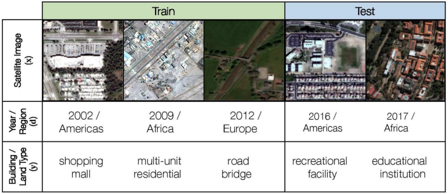
(a) Motion Planning



(b) Audio



(c) Video Streaming



(d) Functional Map (Image)

Feature Selection for Distribution Shift

- Given two batches of data, how should one know how the shift can be explained by the individual features? It can be formulated as a Feature Selection problem for the Distribution Shift(both covariate shift and concept drift).
- It's a problem that every data scientist will encounter with real-world scenarios, tractable practical values as well as the promising extensions build upon the current Statistical & Machine Learning Methodologies.
- This procedure can be embedded in the deployed ML systems - along with the existing practices including KL-divergence & PSI.

Feature Selection for Covariate Shift

We want to understand the difference of $P(\mathbf{X}_{exist})$ and $P(\mathbf{X}_{new})$ - which subset of features are contributing significantly to the covariate shift?

- **MMD-LOCO**: Maximum Mean Discrepancy(MMD) equipped with the leave-one-covariate procedure - calculate the difference as the ranking mechanism for the feature selection.
- **RF-Domain Classifier**: A binary classifier is trained on distinguishing the two batches of datasets $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, borrowing the ideas from the C2ST(Classifier Two-Sample Testing Procedure) and regard the variable importance ranking for the feature selection criterion.
- The random forest based procedure is highly competitive in comparison to the existing SOTA - **FSL-Net**.

Feature Selection for Concept Drift

- Regarding of concept drift, the causal inference methods can be leveraged including **permuCATE(concise feature selection)** & **causal forest split VIMP procedure(most parsimonious, relatively efficient)** are recommended in practice.
- The causal inference method is designed to detect the equivalence of conditional mean $E[Y|X, T = 1], E[Y|X, T = 0]$. The numerical studies and real-data analysis also demonstrated the SOTA performance of the causal inference methods in a wide variety of types of data.
- The software is open-sourced in `https://github.com/ruanhq-StatsML/FSDS-FeatureSelection_for_DistributionShift`.

Production Value

Understanding distribution shift from feature level has direct practical value in the production ML System.

- The two-step procedure for the feature selection can enable the transformation from the hypothesis testing procedure to the feature selection mechanism.
- The covariate shift & concept drift attribution procedure, coupled with other model monitoring practices, can enable the near-real-time, automatic feature removal & granular root-cause diagnosis with the reduced MTTD/MTTA.

Production Value

Integration into the Production System:

- The proposed feature selection procedure for distribution shift can be seamlessly integrated into existing model monitoring pipelines, complementing conventional metrics such as PSI, KL divergence, and the KS statistic in deployed ML systems.
- While fully assumption-less identification of distribution shift drivers is infeasible, the feature selection for distribution shift procedure is still capable of extracting a ranked list of the top-k features contributing significantly to the observed distribution shift, providing actionable references for Business, Operation and Engineering teams.

Production Value

This thesis propose two lines of Statistical Methodologies: Streaming Testing and Batch Testing procedures that is highly generalizable to a wide variety of production scenarios.

- Batch Testing: Parallelization available for Permutation Test, computationally feasible for conducting the test each 5 minutes.
- Production Landing for the Streaming Testing: Near real-time inference & hypothesis testing for the Streaming Observations to trigger Model Refresh Pre-Signals.
- Highly Flexible: all of these methodologies can be adapted to a wide variety of types of datasets & modalities.

Future Directions

Several promising applied research avenues naturally extend our framework, each offering tangible value for production ML systems:

- **Streaming Monitoring & Detection:** Extend to continuous micro-batches using neural encoders (`TarNet/DragonNet`) for online streaming OOD detection.
- **Modality Generalization:** Generalize the feature selection to multi-modal settings (AV, motion planning) beyond tabular data, more importantly the multi-modal information regarding the feature selection and data manipulation are what you will need.

Use Cases

Several promising use cases can be leveraged and tangible built upon on top of the current works and the use cases.

- **Region Selection:** Develop localized sub-population selection strategies for OOD detection and domain adaptation characterization procedures with respect to different regions - VAE on the latent space with the random forest classifier is all you will need.
- **Model Monitoring + Contextual Bandit**: Several Deployed ML model, identify the robust feature subsets to ensure the model continuously generate business impacts - ϵ-Greedy + Enhanced Feature Engineering is probably what you need!